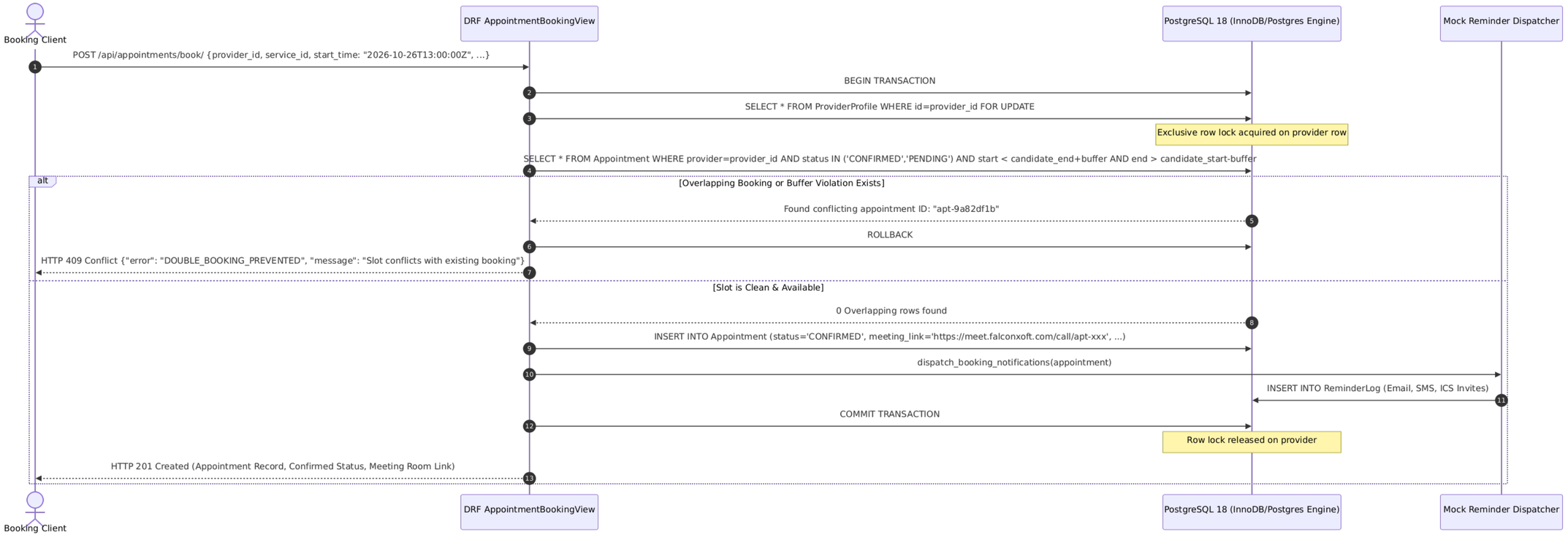

Booking Client
DRF AppointmentBookingView
PostgreSQL 18 (InnoDB/Postgres Engine)
Mock Reminder Dispatcher
POST /api/appointments/book/ {provider_id, service_id, start_time: "2026-10-26T13:00:00Z", ...}
BEGIN TRANSACTION
SELECT * FROM ProviderProfile WHERE id=provider_id FOR UPDATE
Exclusive row lock acquired on provider row
SELECT * FROM Appointment WHERE provider=provider_id AND status IN ('CONFIRMED','PENDING') AND start < candidate_end+buffer AND end > candidate_start-buffer
alt
[Overlapping Booking or Buffer Violation Exists]
Found conflicting appointment ID: "apt-9a82df1b"
ROLLBACK
HTTP 409 Conflict {"error": "DOUBLE_BOOKING_PREVENTED", "message": "Slot conflicts with existing booking"}
[Slot is Clean & Available]
0 Overlapping rows found
INSERT INTO Appointment (status='CONFIRMED', meeting_link='https://meet.falconxoft.com/call/apt-xxx', ...)
dispatch_booking_notifications(appointment)
INSERT INTO ReminderLog (Email, SMS, ICS Invites)
COMMIT TRANSACTION
Row lock released on provider
HTTP 201 Created (Appointment Record, Confirmed Status, Meeting Room Link)
Booking Client
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